

ST GALLEN, Switzerland

WMO: 066810

Lat: 47.4256N

Lon: 9.3986E

Elev: 778

StdP: 92.32

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.0	(c) -8.1	(d) -12.7	(e) 1.4	(f) -6.7	(g) -11.0	(h) 1.6	(i) -5.8	(j) 9.2	(k) 6.3	(l) 8.2	(m) 5.8	(n) 1.3	(o) 30	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.4	(c) 27.1	(d) 18.6	(e) 25.5	(f) 17.9	(g) 24.0	(h) 17.3	(i) 19.7	(j) 25.2	(k) 18.7	(l) 24.0	(m) 17.9	(n) 22.9	(o) 2.5	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.8	23.1	16.7	13.1	21.9	15.9	12.4	20.8	59.9	25.2	56.6	24.1	53.7	22.9	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.3	(b) 6.0	(c) 5.0	(e) -11.9	(f) 29.6	(g) 2.5	(h) 1.5	(i) -13.7	(j) 30.7	(k) -15.2	(l) 31.6	(m) -16.7	(n) 32.5	(o) -18.5	(p) 33.6		
(a) 7.3	(b) 6.0	(c) 5.0	(e) -12.5	(f) 21.6	(g) 2.5	(h) 1.0	(i) -14.3	(j) 22.4	(k) -15.7	(l) 23.0	(m) -17.2	(n) 23.6	(o) -19.0	(p) 24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.1	1.0	4.6	8.4	12.5	16.1	17.8	17.4	13.3	9.6	4.4	1.3		
	DBStd	7.59	4.73	5.03	4.60	4.52	4.02	4.09	3.51	3.49	3.56	4.10	4.51	4.62		
	HDD10.0	1355	307	253	175	83	20	2	0	1	10	58	174	271		
	HDD18.3	3537	564	486	427	299	184	92	53	59	153	272	417	529		
	CDD10.0	960	0	1	7	34	98	184	243	231	110	45	7	1		
	CDD18.3	101	0	0	0	0	5	23	37	32	3	0	0	0		
	CDH23.3	461	0	0	0	1	16	118	185	136	4	0	0	0		
	CDH26.7	65	0	0	0	0	1	16	28	21	0	0	0	0		
(14) Wind	WSAvg	2.0	2.0	2.0	2.3	2.2	2.2	2.0	2.0	1.8	1.7	1.7	1.8	2.0		
(15) Precipitation	PrecAvg	1461	81	88	100	105	147	169	172	165	124	101	109	101		
	PrecMax	1891	183	245	208	186	335	270	324	277	211	206	237	230		
	PrecMin	1144	8	27	36	27	31	72	71	38	59	12	5	10		
	PrecStd	155	48	50	44	42	65	51	61	62	39	44	50	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	13.3	16.7	21.3	25.2	28.7	29.4	29.0	23.9	20.8	17.9	14.7		
		MCWB	6.4	6.4	9.3	12.1	15.8	18.9	19.6	19.1	16.7	12.9	10.0	7.5		
	2%	DB	10.1	11.1	14.4	19.1	23.1	26.3	27.1	26.5	22.0	18.3	15.2	10.9		
		MCWB	5.9	5.5	8.1	11.1	14.8	18.0	18.4	18.9	16.3	12.9	9.1	6.5		
	5%	DB	8.4	9.4	12.8	17.3	21.0	24.6	25.4	24.7	20.3	16.7	12.7	9.3		
		MCWB	4.9	4.7	7.4	10.3	13.8	17.4	18.0	18.0	15.4	12.3	8.6	5.8		
	10%	DB	6.7	7.8	11.1	15.4	19.1	22.7	23.7	23.1	18.7	15.2	10.7	7.7		
		MCWB	3.8	3.9	6.6	9.4	12.9	16.6	17.3	17.3	14.7	11.6	7.6	4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	7.6	10.4	13.2	17.3	20.7	20.9	20.9	18.3	15.3	11.2	8.2		
		MCDB	10.5	11.2	14.9	19.4	23.2	26.6	27.1	26.5	22.4	18.1	15.4	11.4		
	2%	WB	6.5	6.3	9.1	11.9	15.8	19.1	19.5	19.6	17.1	13.9	10.0	7.3		
		MCDB	9.1	9.9	13.1	17.7	21.5	24.5	25.3	25.0	20.8	16.8	13.2	10.4		
	5%	WB	5.2	5.3	8.0	10.8	14.5	18.0	18.6	18.7	16.0	12.9	8.9	6.2		
		MCDB	8.0	8.6	11.7	16.2	19.7	23.3	24.2	23.6	19.5	15.8	12.2	8.9		
	10%	WB	4.0	4.2	7.0	9.7	13.4	17.0	17.7	17.8	15.0	12.0	7.9	5.1		
		MCDB	6.4	7.3	10.6	14.4	18.2	21.9	22.9	22.3	17.9	14.7	10.6	7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	6.2	7.3	7.5	7.4	7.4	6.9	6.2	5.4	4.8	4.6		
		MCDBR	6.3	7.1	8.4	9.5	9.8	9.6	9.5	9.1	8.2	7.5	7.8	6.8		
	5% WB	MCWBR	4.2	4.3	5.0	5.3	5.5	5.5	5.4	5.4	5.0	4.7	4.3	4.1		
		MCWBR	5.8	6.5	7.5	8.9	9.1	8.9	8.8	8.5	7.4	6.5	6.7	6.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.275	0.298	0.335	0.371	0.385	0.392	0.393	0.385	0.358	0.330	0.300	0.271		
		taud	2.431	2.371	2.358	2.309	2.314	2.331	2.320	2.358	2.417	2.484	2.507	2.448		
	Ebn at Noon	Ebn at Noon	822	869	885	881	879	872	865	857	850	824	785	783		
		Edh at Noon	66	87	103	120	124	123	122	113	97	78	62	58		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.17	2.01	3.22	4.33	4.98	5.72	5.54	4.83	3.62	2.28	1.27	0.96		
	RadStd	RadStd	0.13	0.27	0.43	0.62	0.50	0.53	0.51	0.50	0.37	0.25	0.15	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.68	+0.43	+0.36	N/A	N/A	+84	+26
(47) Regional (115 neighbors)		N/A	N/A	N/A	+0.67	+0.42	+0.36	N/A	N/A	+54	+13

Nomenclature: See separate page